



Level 2

Challenge D



Directions:

In the `prolog_challenges.pl` file, create a predicate called `"minheap"` that does the following:

The predicate will take in a list of integers and will be true if the list represents a min heap or will fail if not. Recall that a min heap is normally represented as an array with the root being at index 0. The predicates defined in `library.pl` will help you and are already included in your challenges file. It is strongly recommended to look over the file and understand what all it does.

You will need to create `minheap/1` that takes in a list as well as `minheap/2` that takes a list and an index. `minheap/1` should invoke `minheap/2` sending zero as the index (i.e. the root index). `minheap/2` should check if the index is out of bounds and allow that index if so (do not fail). If index is in bounds, it should check for min heap violations in the child values, then recursively check the child indices. It should fail if any violations are found. Remember that you can use `fail/0` as a predicate that always fails (i.e. once you find a value in the list that doesn't follow min heap rules, you can fail immediately).

There are 9 lists imported from `library.pl` to help you check your `minheap` predicate.

Examples:

```
/* query */ ?- list1(List), minheap(List).
/* result */ true.

/* query */ ?- list2(List), minheap(List).
/* result */ true.

/* query */ ?- list3(List), minheap(List).
/* result */ true.

/* query */ ?- list4(List), minheap(List).
/* result */ true.

/* query */ ?- list5(List), minheap(List).
/* result */ true.

/* query */ ?- list6(List), minheap(List).
/* result */ true.

/* query */ ?- list7(List), minheap(List).
/* result */ false.

/* query */ ?- list8(List), minheap(List).
/* result */ false.

/* query */ ?- list9(List), minheap(List).
/* result */ false.
```